

Machine Learning and Pattern Recognition — Lecture 01: Introduction and Nearest Neighbors

المحاضرة الأولى: مقدمة في تعلم الآلة وخوارزمية أقرب الجيران (kNN)

Part 1: Multiple Choice Questions (25)

1. According to Tom Mitchell's definition, a computer program is said to learn from experience E with respect to tasks T and performance measure P if:

حسب تعريف Tom Mitchell، البرنامج يتعلم من الخبرة E لما:

- A) its source code is rewritten by the programmer each time new data arrives
- B) its performance at tasks in T, as measured by P, improves with experience E**
- C) it memorizes every training example exactly and reproduces it on request
- D) it follows a fixed set of rules that the programmer wrote before seeing any data

الإجابة الصحيحة: B) its performance at tasks in T, as measured by P, improves with experience E
ليه صح: التعريف يربط التعلم بتحسّن الأداء P على المهام T مع زيادة الخبرة E. الحفظ الحرفي أو القواعد الثابتة مش بتمثل تحسّن مع الخبرة، وطول الكود ملوش علاقة بالأداء.

Ref: Lecture 01 — Slide 5

2. Which of the following is given in the lecture as a reason to use a learning algorithm?

أي من التالي مذكور في المحاضرة كسبب لاستخدام خوارزمية تعلّم؟

- A) The correct behavior can easily be programmed by hand using a few rules
- B) Labeled data is never needed because the system ignores experience
- C) The goal is to avoid using any data and rely on fixed formulas
- D) The system must adapt to a changing environment, such as spam detection**

الإجابة الصحيحة: D) The system must adapt to a changing environment, such as spam detection
ليه صح: من الأسباب المذكورة: صعوبة كتابة الحل يدويًا، وحاجة النظام للتكيّف مع بيئة متغيرة (مثل spam detection). لو الحل سهل يدويًا فمفيش داعي للتعلم، والتعلم محتاج بيانات أو خبرة.

Ref: Lecture 01 — Slide 6

3. In which type of machine learning does an agent interact with the world and learn to maximize a scalar reward signal?

في أي نوع من تعلم الآلة يتفاعل الـ agent مع العالم ويتعلم يعظّم إشارة مكافأة رقمية؟

- A) Reinforcement learning**
- B) Supervised learning
- C) Unsupervised learning
- D) Instance-based regression

الإجابة الصحيحة: A) Reinforcement learning
ليه صح: الـ reinforcement learning هو اللي فيه agent وreward. الـ supervised محتاج labels صحيحة، والـ unsupervised مفيهوش labels ويدور على أنماط.

Ref: Lecture 01 — Slide 7

4. In supervised learning, which is the training set composed of?

في ال supervised learning، مكوّنة من إيه مجموعة التدريب؟

- A) Inputs only, with no labels
- B) Only reward signals received from an environment
- C) Only the hyperparameters of the model

D) Inputs together with their corresponding labels

الإجابة الصحيحة: D) Inputs together with their corresponding labels
ليه صح: ال supervised learning معناه إن عندنا مدخلات ومعها labels (الإجابة الصحيحة). المدخلات بدون labels هي حالة unsupervised، وال rewards هي reinforcement.

Ref: Lecture 01 — Slide 16

5. In a regression problem, the target t is:

في مسألة ال regression، ال t target بيكون:

- A) an element of a discrete set $\{1, \dots, C\}$
- B) always a binary value 0 or 1
- C) always a vector of pixel values

D) a real number, for example a stock price

الإجابة الصحيحة: D) a real number, for example a stock price
ليه صح: في ال regression ال t رقم حقيقي (زي سعر سهم). العنصر من مجموعة منفصلة $\{C..1\}$ ده تعريف ال classification.

Ref: Lecture 01 — Slide 20

6. Why is representing the input as a vector in $\mathbb{R}^d$ a good strategy?

ليه تمثيل المدخلات كمتجه في $\mathbb{R}^d$ استراتيجية كويسة؟

- A) Because it removes the need for any labels in supervised problems

B) Because vectors are easy to manipulate and we can apply linear algebra

- C) Because it always reduces every data type to two dimensions
- D) Because it guarantees zero error on the training set for any model

الإجابة الصحيحة: B) Because vectors are easy to manipulate and we can apply linear algebra
ليه صح: المحاضرة بتقول إن المتجهات تمثيل ممتاز لأننا نقدر نستخدم الجبر الخطي. التمثيل ميلغيش ال labels ولا بيضمن خطأ صفر.

Ref: Lecture 01 — Slide 18

7. Why does the lecture recommend vectorizing computations (for example with NumPy)?

ليه المحاضرة بتنصح بعمل vectorization للحسابات (مثلاً بـ NumPy)؟

A) Expressing them as matrix/vector operations exploits hardware efficiency and makes code cleaner

- B) It removes the need to derive the algorithm mathematically before coding
- C) It makes the training data larger so the model fits it better
- D) It is required for the model to have labels in supervised learning

الإجابة الصحيحة: A) Expressing them as matrix/vector operations exploits hardware efficiency and makes code cleaner
ليه صح: صياغة الحسابات كعمليات matrix/vector بتستغل كفاءة الهاردوير وتخلي الكود أنصف وأسهل في القراءة. ملهاش علاقة بحجم البيانات ولا بال labels.

Ref: Lecture 01 — Slide 14

8. What is the basic idea of the nearest neighbor method for classifying a novel input x ?

إيه الفكرة الأساسية لطريقة أقرب جار (nearest neighbor) لتصنيف مدخل جديد x ؟

- A) Fit a straight line through all training points
- B) Average the labels of all training examples
- C) Find the nearest training input to x and copy its label**
- D) Build a tree by splitting on features

الإجابة الصحيحة: C) Find the nearest training input to x and copy its label

ليه صح: الفكرة إننا نلاقي أقرب مدخل في الـ training set وننسخ الـ label بتاعه. الخط المستقيم هو regression، والشجرة هي decision tree، والمتوسط الكلي بيتجاهل القرب.

Ref: Lecture 01 — Slide 21

9. How does the lecture formalize 'nearest' in nearest neighbors?

إزاي المحاضرة بتعرّف كلمة 'أقرب' في nearest neighbors؟

- A) Using Euclidean distance, although other choices are possible**
- B) Using Manhattan distance only, with no other choices allowed
- C) Using Hamming distance only, with no other choices allowed
- D) Using the number of shared labels between two examples

الإجابة الصحيحة: A) Using Euclidean distance, although other choices are possible

ليه صح: المحاضرة بتستخدم المسافة الإقليدية (Euclidean) وتذكر إن في اختيارات تانية ممكنة، فمقياس مسافة واحدة بس مفروضة (Manhattan أو Hamming).

Ref: Lecture 01 — Slide 21

10. What is a decision boundary in a classification task?

إيه هو الـ decision boundary في مسألة تصنيف؟

- A) A line or surface that divides different groups in the classification task**
- B) The training example that has the largest numeric label value
- C) The set of all misclassified test examples in the dataset
- D) The validation set that is used to tune the value of k

الإجابة الصحيحة: A) A line or surface that divides different groups in the classification task

ليه صح: الـ decision boundary خط أو سطح يفصل المجموعات (classes) المختلفة ويوضح أي منطقة تنتمي لأي class.

Ref: Lecture 01 — Slide 22

11. What is a main weakness of the 1-nearest-neighbor classifier?

إيه نقطة الضعف الأساسية في 1-nearest-neighbor؟

- A) It cannot be applied to numeric features such as pixel values
- B) It requires a very long training phase before predicting
- C) It is sensitive to noise or mislabeled data (class noise)**
- D) It always underfits the data regardless of the labels

الإجابة الصحيحة: C) It is sensitive to noise or mislabeled data (class noise)

ليه صح: لأنه بينسخ label أقرب نقطة بس، فلو النقطة دي غلط أو فيها noise التصنيف هيغلط. وهو غالبًا بيعمل overfit مش underfit، ومفيهوش مرحلة تدريب طويلة.

Ref: Lecture 01 — Slide 23

12. What is the proposed solution to the noise sensitivity of nearest neighbors?

إيه الحل المقترح لحساسية nearest neighbors لل noise؟

A) Smooth the decision by having the k nearest neighbors vote

- B) Use only the single nearest neighbor with a larger weight
- C) Remove all training points that are far from x
- D) Double the number of dimensions

الإجابة الصحيحة: A) Smooth the decision by having the k nearest neighbors vote
ليه صح: الحل إن k جيران يصوتوا فينعمل smoothing، فتأثير نقطة واحدة غلط بيقل. الاعتماد على جار واحد بس هو أصل المشكلة.

Ref: Lecture 01 — Slide 24

13. What is the classification output of the kNN algorithm?

إيه ناتج التصنيف في خوارزمية kNN؟

A) The majority class among the k examples closest to the test instance

- B) The class of the farthest training example in the set
- C) The mean of the k distances to the test instance
- D) The class that has the fewest examples among the neighbors

الإجابة الصحيحة: A) The majority class among the k examples closest to the test instance
ليه صح: الخوارزمية بتلاقي k أمثلة الأقرب لل test instance وتطلع ال majority class بينهم.

Ref: Lecture 01 — Slide 25

14. What is the risk of choosing a very small k?

إيه خطر اختيار k صغيرة جدًا؟

- A) Underfitting: the model averages over too many examples
- B) The algorithm cannot produce any prediction for new inputs
- C) Overfitting: it becomes sensitive to random abnormalities in the training data**
- D) The memory requirement becomes zero because nothing is stored

الإجابة الصحيحة: C) Overfitting: it becomes sensitive to random abnormalities in the training data
ليه صح: k الصغيرة بتلقط الأنماط الدقيقة لكن ممكن تعمل overfit وتتأثر بالقيم الشاذة العشوائية. ال underfitting هو مشكلة k الكبيرة.

Ref: Lecture 01 — Slide 31

15. What is the risk of choosing a very large k?

إيه خطر اختيار k كبيرة جدًا؟

- A) Overfitting to random abnormalities in the training data
- B) Underfitting: it may fail to capture important regularities**
- C) The model becomes sensitive to every single noisy point
- D) The algorithm needs the labels to be removed from the data

الإجابة الصحيحة: B) Underfitting: it may fail to capture important regularities
ليه صح: k الكبيرة بتخلي التنبؤ مستقر لأنه بيتوسط على أمثلة كثير، لكن ممكن تفوت أنماط مهمة فيحصل underfitting.

Ref: Lecture 01 — Slide 31

16. Compared with $k = 1$, the decision boundary of kNN with $k = 15$ (shown in the figures) is:

مقارنةً بـ $k = 1$ ، ال decision boundary في kNN لما $k = 15$ سيكون:

- A) more jagged and more complex
- B) exactly identical
- C) undefined for two classes
- D) smoother and less complex**

الإجابة الصحيحة: D) smoother and less complex

ليه صح: زيادة k بتعمل averaging على جيران أكثر فالحدود بتبقى أنعم وأبسط. $k = 1$ بيدي حدود متعرجة بتتبع كل نقطة.

Ref: Lecture 01 — Slides 26-27 and 31

17. How can we measure the generalization error of a model?

إزاي نقيس generalization error لنموذج؟

- A) By computing the error rate on the training set only
- B) By counting the number of hyperparameters in the model
- C) By measuring the error rate on new examples (a test set)**
- D) By measuring the training time of the algorithm

الإجابة الصحيحة: C) By measuring the error rate on new examples (a test set)

ليه صح: ال generalization error هو نسبة الخطأ على أمثلة جديدة لم يشوفها النموذج، وينقيسه بـ test set. الخطأ على ال training set ممكن يكون صغير جدًا من غير ما النموذج بعمم.

Ref: Lecture 01 — Slide 32

18. Why is k in kNN called a hyperparameter?

ليه k في kNN بتتسمى hyperparameter؟

- A) It is learned automatically by gradient descent during training
- B) It cannot be fit as part of the learning algorithm itself**
- C) It is the label of the nearest training point
- D) It is one of the input features of each example

الإجابة الصحيحة: B) It cannot be fit as part of the learning algorithm itself

ليه صح: ال hyperparameter قيمة مبنقدش نتعلمها كجزء من خوارزمية التعلم نفسها، فينبسطها بـ validation set. مش feature ولا label.

Ref: Lecture 01 — Slide 33

19. What is the validation set used for?

ال validation set يُستخدم في إيه؟

- A) Tuning hyperparameters such as k**
- B) Measuring the final generalization performance
- C) Storing the training data in memory
- D) Computing the Euclidean distances

الإجابة الصحيحة: A) Tuning hyperparameters such as k

ليه صح: ال validation set بنضبط بيه ال hyperparameters زي k . أما قياس الأداء النهائي فده دور ال test set اللي يُستخدم في الآخر بس.

Ref: Lecture 01 — Slide 33

20. What happens to the volume of the feature space as the number of dimensions increases?

إيه اللي بيحصل لحجم فراغ ال features لما عدد الأبعاد يزيد؟

A) It shrinks, so the data becomes denser

B) It grows exponentially, so the data becomes sparse

C) It stays constant while the points move closer

D) It grows linearly and the data stays dense

الإجابة الصحيحة: B) It grows exponentially, so the data becomes sparse

ليه صح: الحجم بيزيد أُسِّيًّا فالبيانات بتبقى متباعدة (sparse). ده أساس ال curse of dimensionality وبيأثر على kNN.

Ref: Lecture 01 — Slide 34

21. Why does high dimensionality hurt nearest neighbors?

ليه الأبعاد العالية بتضر nearest neighbors؟

A) Distances become exactly zero between all pairs of points in the set

B) The nearest neighbor always turns out to be the farthest point

C) The labels of the training points disappear from the dataset

D) All points appear roughly equidistant, so picking a label is hard

الإجابة الصحيحة: D) All points appear roughly equidistant, so picking a label is hard

ليه صح: في الأبعاد العالية المسافات بين أي نقطتين بتقارب بعض فالمقارنة بالقرب تفقد معناها. المسافات مش بتبقى صفر ولا بتتعاكس.

Ref: Lecture 01 — Slide 36

22. What is the simple fix proposed for the sensitivity of nearest neighbors to feature ranges?

إيه الحل البسيط المقترح لحساسية nearest neighbors لمدى قيم ال features؟

A) Set k equal to the total number of training examples available

B) Delete every feature except the first one in the vector

C) Normalize each dimension to have zero mean and unit variance

D) Use only binary labels for all of the examples

الإجابة الصحيحة: C) Normalize each dimension to have zero mean and unit variance

ليه صح: بنحسب mean و standard deviation لكل بُعد ونعمل $\tilde{x} = (x - \mu) / \sigma$. المحاضرة بتحدّر إن في بعض المسائل ال scale بيكون مهم.

Ref: Lecture 01 — Slide 37

23. How many computations does naive kNN need at training time?

كام عملية حسابية بتحتاجها kNN البسيطة وقت التدريب؟

A) $O(ND)$ for computing all pairwise distances

B) Zero, since it only stores the data

C) $O(N \log N)$ for sorting the training set

D) $O(N^2)$ for building a distance matrix

الإجابة الصحيحة: B) Zero, since it only stores the data

ليه صح: kNN بتخزن البيانات بس، فتكلفة التدريب صفر. التكلفة كلها وقت الاختبار لكل $O(ND)$ query: للمسافات و $O(N \log N)$ للترتيب.

Ref: Lecture 01 — Slide 38

24. In the digit classification example, what error did KNN with shape contexts achieve on MNIST compared with Euclidean KNN?

في مثال تصنيف الأرقام، إيه نسبة الخطأ اللي حققتها KNN باستخدام shape contexts على MNIST مقارنةً بـ Euclidean KNN؟

A) 3% error versus 0.63% error

B) 0% error versus 3% error

C) 10% error versus 5% error

D) 0.63% error versus 3% error

الإجابة الصحيحة: D) 0.63% error versus 3% error

ليه صح: ال shape contexts حققت 0.63% مقابل 3% لـ Euclidean KNN، وده بيوضح إن مقياس تشابه كويس بيحسّن kNN كتير. الاختيار العكسي بيقلب الأرقام.

Ref: Lecture 01 — Slide 40

25. What was special about the '80 Million Tiny Images' dataset?

إيه المميز في dataset اسمها '80 Million Tiny Images'؟

A) It was a small dataset of high-resolution medical images for diagnosis

B) It contained only audio waveforms recorded from speakers

C) The first extremely large image dataset: color images scaled down to 32 x 32

D) It was a dataset of only 80 labeled images used for testing

الإجابة الصحيحة: C) The first extremely large image dataset: color images scaled down to 32 x 32

ليه صح: كانت أول dataset ضخمة جدًا للصور، مكوّنة من صور ملونة مصغّرة لـ 32 x 32، ومع البيانات الكبيرة بتلاقي matches دلالية أحسن (مع مقياس تشابه مختار بعناية).

Ref: Lecture 01 — Slide 41

Part 2: Essay and Calculation Questions (15)

1. Define machine learning and explain why we may want to use a learning algorithm instead of hand-written rules.

عرّف تعلم الآلة واطرح ليه ممكن نستخدم خوارزمية تعلّم بدل القواعد المكتوبة يدويًا.

Model answer:

Machine learning: programming an algorithm to automatically learn from data or from experience. Tom Mitchell: a program learns from experience E for tasks T with performance P if its performance on T, measured by P, improves with E.

Reasons to use learning:

- Hard to code a solution by hand (e.g., vision, speech).
- The system must adapt to a changing environment (e.g., spam detection).
- We want the system to perform better than the human programmers.
- Privacy/fairness (e.g., ranking search results).

Ref: Lecture 01 — Slides 5-6

2. Compare supervised, unsupervised and reinforcement learning, and mention application areas of machine learning from the lecture.

قارن بين supervised وunsupervised وreinforcement learning واذكر مجالات تطبيق ML من المحاضرة.

Model answer:

Supervised learning: we have labeled examples of the correct behavior.

Unsupervised learning: no labeled examples; we look for interesting patterns in the data.

Reinforcement learning: an agent interacts with the world and learns to maximize a scalar reward signal.

Applications: computer vision (object detection, semantic segmentation, pose estimation), natural language processing (machine translation, sentiment analysis, spam filtering, chatbots), and e-commerce/recommender systems (Amazon, Netflix).

Ref: Lecture 01 — Slides 7-10

3. Describe the supervised learning setup: inputs, labels, input vectors, and the difference between regression and classification.

اشرح إعداد ال supervised learning: المدخلات وال labels وال input vectors والفرق بين regression وclassification.

Model answer:

We are given a training set of inputs and corresponding labels (e.g., image $\rightarrow$ object category, audio $\rightarrow$ text).

Inputs of any type (images, text, audio, transactions) are represented as an input vector x in $\mathbb{R}^d$; vectors allow linear algebra. We can use raw pixels, but a vector of meaningful features is much better.

The training set is $\{(x(1), t(1)), \dots, (x(N), t(N))\}$.

Regression: t is a real number (e.g., stock price).

Classification: t is an element of a discrete set $\{1, \dots, C\}$.

Ref: Lecture 01 — Slides 16-20

4. Explain the nearest neighbor and k-nearest neighbors algorithms, including how the kNN prediction is made.

اشرح خوارزميتي nearest neighbor وk-nearest neighbors وإزاي بيتعمل التنبؤ في kNN.

Model answer:

Nearest neighbor: given a novel input x , find the nearest input vector in the training set D_{tr} and copy its label.

'Nearest' is formalized with Euclidean distance (other choices possible).

Problem: it is sensitive to noise or mislabeled data (class noise).

Solution (kNN): smooth by letting the k nearest neighbors vote.

Algorithm: (1) find the k examples in D_{tr} closest to the test instance x ; (2) the classification output is the majority class among them.

Ref: Lecture 01 — Slides 21-25

5. Discuss the trade-off in choosing k in kNN, and what the decision boundaries for $k = 1$ and $k = 15$ show.

ناقش ال trade-off في اختيار k في kNN وإيه اللي بتوضحه حدود القرار لـ $k = 1$ و $k = 15$.

Model answer:

Small k : good at capturing fine-grained patterns, but may overfit, i.e., be sensitive to random abnormalities in the training data. The boundary for $k = 1$ follows individual points and is jagged.

Large k : makes stable predictions by averaging over many examples, but may underfit, i.e., fail to capture important regularities. The boundary for $k = 15$ is smoother.

The optimal k depends on the number of data points n . We control the complexity of the model by varying k .

Ref: Lecture 01 — Slides 26-27 and 31

6. What is a hyperparameter? Explain the roles of the validation set and the test set and how generalization is measured.

إليه هو الـ hyperparameter؟ اشرح دور الـ validation set والـ test set وإزاي بنقيس الـ generalization.

Model answer:

We want the algorithm to generalize to data it has not seen before. Generalization error = error rate on new examples, measured with a test set.

k is a hyperparameter: something we cannot fit as part of the learning algorithm itself.

We tune hyperparameters using a validation set.

The test set is used only at the very end, to measure the generalization performance of the final configuration.

Ref: Lecture 01 — Slides 32-33

7. Explain the curse of dimensionality and its impact on the performance of kNN.

اشرح الـ curse of dimensionality وتأثيره على أداء kNN.

Model answer:

The curse of dimensionality refers to phenomena that arise with high-dimensional data. As the number of dimensions increases, the volume of the space grows exponentially, so the data becomes sparse (points are farther apart).

Consequences: increased computational complexity, overfitting, and deteriorating performance of some algorithms.

For kNN: it assumes that points close in input space have close (the same) outputs. In high dimensions, distances become more uniform and all points appear roughly equidistant, so the notion of distance loses meaning and it is hard to select the label for a test point.

Ref: Lecture 01 — Slides 34-36

8. Why and how do we normalize features in nearest neighbors? Is normalization always appropriate?

ليه وإزاي بنعمل normalization للـ features في nearest neighbors؟ وهل دايماً مناسب؟

Model answer:

Nearest neighbors is sensitive to the ranges of different features; often the units are arbitrary, so a feature with a large range dominates the distance.

Fix: normalize each dimension to zero mean and unit variance. Compute the mean μ_j and standard deviation σ_j , and take $x_{\sim j} = (x_j - \mu_j) / \sigma_j$.

Caution: depending on the problem, the scale might be important, so normalization is not always appropriate.

Ref: Lecture 01 — Slide 37

9. Analyze the computational cost of naive kNN at training time and test time, and mention its memory requirement.

حلّل تكلفة الحساب في kNN البسيطة وقت التدريب ووقت الاختبار وإذكر متطلبات الذاكرة.

Model answer:

Training time: 0 computations (it only stores the data).

Test time, per query (naive algorithm): compute D-dimensional Euclidean distances to N data points: $O(ND)$; then sort the distances: $O(N \log N)$.

This is repeated for every query, which is very expensive by the standards of a learning algorithm. It also needs to store the entire dataset in memory.

Much work has gone into algorithms and data structures for efficient nearest neighbors in high dimensions or with large datasets.

Ref: Lecture 01 — Slide 38

10. Why does the lecture derive algorithms by hand although frameworks such as PyTorch and TensorFlow do much of the work? Summarize the conclusions about kNN.

ليه المحاضرة بتشتق الخوارزميات يدويًا رغم إن frameworks زي PyTorch و TensorFlow بتعمل شغل كثير؟ ولخص الخلاصة عن kNN.

Model answer:

Frameworks (PyTorch, TensorFlow, JAX) provide automatic differentiation, computation graphs, algorithm libraries and GPU support. We still derive things by hand so we know what to do if something goes wrong: debugging learning algorithms requires understanding what goes on beneath the hood.

Conclusions on kNN: a simple algorithm that does all its work at test time (in a sense, no learning); the complexity can be controlled by varying k ; it suffers from the curse of dimensionality. The next topic is parametric models, which learn a compact summary of the data.

Ref: Lecture 01 — Slides 15 and 43

11. Calculation: Compute the Euclidean distance between the points $p = (1, 2, 3)$ and $q = (4, 6, 3)$.

مسألة حسابية: احسب المسافة الإقليدية بين النقطتين $p = (1, 2, 3)$ و $q = (4, 6, 3)$.

Model answer:

Formula: $d(p, q) = \sqrt{\sum_i (p_i - q_i)^2}$.

Differences: $(1-4) = -3$, $(2-6) = -4$, $(3-3) = 0$.

Squares: $9 + 16 + 0 = 25$.

$d = \sqrt{25} = 5$.

Ref: Lecture 01 — Slide 21

12. Calculation: Using the first example in the lecture, $x = [4, 5, 10, 4, 3, 11, 14, 9, 8, 10, 12]$, $y = [21, 19, 24, 17, 16, 25, 24, 23, 22, 21, 21]$, classes = $[0, 0, 1, 0, 0, 1, 1, 1, 0, 1, 1]$, classify the new point $(8, 21)$ with $k = 1$, $k = 3$ and $k = 5$.

مسألة حسابية: باستخدام المثال الأول في المحاضرة، صنف النقطة الجديدة $(8, 21)$ بـ $k = 1$ و $k = 3$ و $k = 5$.

Model answer:

Euclidean distances from $(8, 21)$, sorted from nearest:

$(8, 22)$ class 0: $d = 1.00$

$(10, 21)$ class 1: $d = 2.00$

$(9, 23)$ class 1: $d = \sqrt{1 + 4} = 2.24$

$(5, 19)$ class 0: $d = \sqrt{9 + 4} = 3.61$

$(10, 24)$ class 1: $d = \sqrt{4 + 9} = 3.61$

$(4, 21)$ class 0 and $(12, 21)$ class 1: $d = 4.00$

$k = 1$: nearest neighbor is class 0, so predicted class = 0.

$k = 3$: neighbors are classes 0, 1, 1, majority = 1.

$k = 5$: neighbors are classes 0, 1, 1, 0, 1 (three of class 1 versus two of class 0), majority = 1.

Note: the prediction changes from 0 to 1 as k grows, which shows the effect of k .

Ref: Lecture 01 — Slide 28

13. Calculation: Using the second example in the lecture (heights x and weights y), classify the new point (170, 55) with k = 3.

مسألة حسابية: باستخدام المثال الثاني في المحاضرة، صنف النقطة الجديدة (170, 55) بـ $k = 3$.

Model answer:

Data: $x = [167, 182, 176, 173, 172, 174, 169, 173, 170]$, $y = [51, 62, 69, 64, 65, 56, 58, 57, 55]$, classes = [0, 1, 1, 1, 1, 0, 1, 1, 1].

Distances to (170, 55): (170,55): 0 (class 1); (169,58): $\sqrt{1+9} = 3.16$ (class 1); (173,57): $\sqrt{9+4} = 3.61$ (class 1); (174,56): $\sqrt{16+1} = 4.12$ (class 0); (167,51): 5.00 (class 0).

3 nearest = class 1, class 1, class 1.

Majority vote: predicted class = 1.

Ref: Lecture 01 — Slide 30

14. Calculation: Normalize the feature values [2, 4, 6, 8] to zero mean and unit variance (use the population standard deviation).

مسألة حسابية: اعمل normalization للقيم [2, 4, 6, 8] لتبقى zero mean و unit variance (استخدم الانحراف المعياري للمجتمع).

Model answer:

Formula: $\tilde{x} = (x - \mu) / \sigma$.

$\mu = (2 + 4 + 6 + 8) / 4 = 5$.

Variance = $((-3)^2 + (-1)^2 + 1^2 + 3^2) / 4 = 20 / 4 = 5$, so $\sigma = \sqrt{5} = 2.236$.

Normalized values: $(2-5)/2.236 = -1.342$; $(4-5)/2.236 = -0.447$; $(6-5)/2.236 = 0.447$; $(8-5)/2.236 = 1.342$.

Ref: Lecture 01 — Slide 37

15. Calculation: A naive kNN classifier has N = 10,000 training points with D = 50 features. Estimate the number of operations for one query and for 200 queries, and the training cost.

مسألة حسابية: مصنف kNN بسيط عنده $N = 10,000$ نقطة تدريب و $D = 50$ feature. قُدّر عدد العمليات لـ query واحد ولـ 200 query وتكلفة التدريب.

Model answer:

Training cost: 0 (only stores the data).

Distances per query: $O(ND) = 10,000 \times 50 = 500,000$.

Sorting per query: $O(N \log N) = 10,000 \times \log_2(10,000) = 10,000 \times 13.29 \approx 132,877$.

Total per query $\approx 500,000 + 132,877 \approx 632,877$ operations.

For 200 queries: $200 \times 632,877 \approx 1.27 \times 10^8$ operations.

The entire dataset (10,000 x 50 values) must also be kept in memory.

Ref: Lecture 01 — Slide 38